



JAIDEV EDUCATION SOCIETY'S
J D COLLEGE OF ENGINEERING AND MANAGEMENT
KATOL ROAD, NAGPUR
Affiliated to DBATU, RTMNU & MSBTE, Mumbai
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An Autonomous Institute, with NAAC "A" Grade
Department of Computer Science & Engineering
"A place to learn; A chance to grow"
Session: 2025-26

MID SEMESTER EXAMINATION-02

Course: B. Tech. in Computer Science & Engineering
Subject: Machine Learning

Year/Semester: 3rd/ 5th Sem

Subject Code: CS5T002

Max Marks:20

Date: 30/09/2025

Duration:- 1 Hr.

Instructions to the Students:

1. All the questions carry marks as indicate.
2. Assume suitable data whenever necessary.
3. Illustrate your answer whenever necessary with the help of neat sketches.

Students should be able to:

1. Understand the basics of machine learning and different types of learning methods.
2. Apply concept learning and clustering algorithms to solve learning problems.
3. Analyze Bayesian learning approaches and learning theories.
4. Design predictive models using regression, kernel methods, and ensemble techniques.
5. Evaluate and compare machine learning algorithms using decision trees and hypothesis testing.

SECTION A

1. All Questions are compulsory.
2. Answer in one word/sentence.

Q. No.	Questions	Level/CO Mapped	Marks
Q.1	Recall the definition of Bayes' Theorem.	L1/CO3	1
Q.2	Tell about what is meant by Hypothesis Space Search in a Decision Tree.	L1/CO5	1
Q.3	State the basic Concept of Decision Tree.	L1/CO5	1
Q.4	Define following terms a) Root Node b) Leaf Node	L1/CO5	1

SECTION B

1. All Questions are compulsory.
2. Solve answer in brief (two, three Sentence)

Q.5	Explain the concept of the Bayes Optimal Classifier and illustrate it using its formula.	L2/CO3	2
Q.6	Outline the main steps followed in constructing a Decision Tree algorithm.	L1/CO5	2
Q.7	In a Certain Population, the Probability that a person has Dengue is $P(D)=1/30000$. Dengue Fever Cusses Neck Pain 80% of the time. The Probability that a person experiences neck pain from any cause is $P(N)=0.02$ what is the probability that a person has dengue given that they have neck pain? Use Bayes' theorem to calculate.	L3/CO3	2
Q.8	Discuss the following Terms a) Simple Random Sampling b) Systematic Sampling.	L2/CO1	2